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LAPORAN PRAKTIKUM

Intrusion Detection System (IDS)

I. TUJUAN PRAKTIKUM

1. Memahami konsep Intrusion Detection System (IDS)
2. Mengimplementasikan IDS sederhana pada sistem Linux
3. Menganalisis hasil deteksi serangan
4. Membandingkan metode signature-based dan anomaly-based IDS

II. DASAR TEORI

Intrusion Detection System (IDS) adalah sistem yang digunakan untuk mendeteksi aktivitas mencurigakan dalam jaringan atau sistem komputer.

Jenis IDS:

- Signature-based IDS
Mendeteksi serangan berdasarkan pola (signature) yang sudah dikenal
- Anomaly-based IDS
Mendeteksi aktivitas yang menyimpang dari perilaku normal

Perbedaan IDS dan IPS:

- IDS → hanya mendeteksi dan memberikan alert
- IPS → mendeteksi dan langsung melakukan pencegahan

III. ALAT DAN BAHAN

- Virtual Machine Linux (Ubuntu)
- Attacker VM ---> Target VM + Snort IDS
- Tools:
 - Snort (IDS)

- Jaringan: local / virtual lab

IV. LANGKAH PRAKTIKUM

Terminal Target VM

1. Instalasi IDS (Snort)

sudo apt update

```
root@kevin:/home/kevin# sudo apt update
Hit:1 http://archive.ubuntu.com/ubuntu focal InRelease
Hit:2 http://archive.ubuntu.com/ubuntu focal-updates InRelease
Hit:3 http://archive.ubuntu.com/ubuntu focal-backports InRelease
Hit:4 http://archive.ubuntu.com/ubuntu focal-security InRelease
Reading package lists... Done
Building dependency tree
Reading state information... Done
All packages are up to date.
root@kevin:/home/kevin#
```

cek ip a

```
root@kevin:/home/kevin# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:2f:19:ae brd ff:ff:ff:ff:ff:ff
    inet 192.168.1.36/24 metric 100 brd 192.168.1.255 scope global dynamic ens33
        valid_lft 171460sec preferred_lft 171460sec
    inet6 fe80::20c:29ff:fe2f:19ae/64 scope link
        valid_lft forever preferred_lft forever
3: ens34: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:2f:19:b8 brd ff:ff:ff:ff:ff:ff
    inet 192.168.237.128/24 metric 100 brd 192.168.237.255 scope global dynamic ens34
        valid_lft 1360sec preferred_lft 1360sec
    inet6 fe80::20c:29ff:fe2f:19b8/64 scope link
        valid_lft forever preferred_lft forever
root@kevin:/home/kevin#
```

- ens33 192.168.1.0/24 (sesuaikan ip anda)

sudo apt install snort -y

Package configuration

Configuring snort

This value is usually "eth0", but this may be inappropriate in some network environments; for a dialup connection "ppp0" might be more appropriate (see the output of "/sbin/ifconfig").

Typically, this is the same interface as the "default route" is on. You can determine which interface is used for this by running "/sbin/route -n" (look for "0.0.0.0").

It is also not uncommon to use an interface with no IP address configured in promiscuous mode. For such cases, select the interface in this system that is physically connected to the network that should be inspected, enable promiscuous mode later on and make sure that the network traffic is sent to this interface (either connected to a "port mirroring/spanning" port in a switch, to a hub, or to a tap).

You can configure multiple interfaces, just by adding more than one interface name separated by spaces. Each interface can have its own specific configuration.

Interface(s) which Snort should listen on:

ens33

<Ok>

Package configuration

Configuring snort

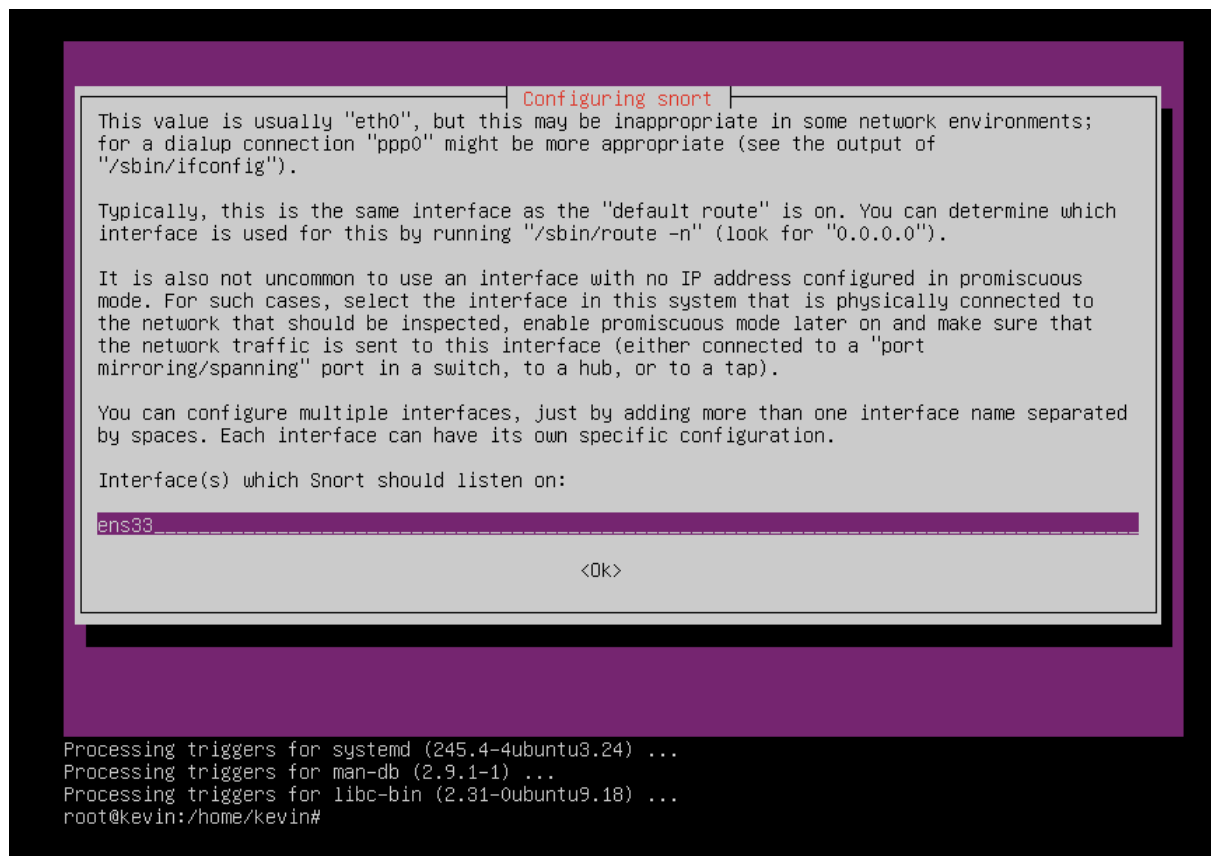
Please use the CIDR form - for example, 192.168.1.0/24 for a block of 256 addresses or 192.168.1.42/32 for just one. Multiple values should be comma-separated (without spaces).

Please note that if Snort is configured to use multiple interfaces, it will use this value as the HOME_NET definition for all of them.

Address range for the local network:

192.168.4.0/24

<Ok>



NOTE : Jika Error pada saat pembuatan file snort.conf Coba Ganti Repositori ke Indonesia (saya menggunakan kartolo data utama) atau buat file snort kosong dulu dengan command dibawah

(wajib dilakukan untuk langkah kedua konfigurasi snort)

```
sudo mkdir -p /etc/snort/rules
```

```
sudo touch /etc/snort/snort.conf
```

```
sudo touch /etc/snort/rules/local.rules
```

2. Konfigurasi Snort

Edit file konfigurasi:

```
sudo nano /etc/snort/snort.conf
```

```
GNU nano 4.8 /etc/snort/snort.conf
# Konfigurasi Jaringan
ipvar HOME_NET 192.168.4.0/24
ipvar EXTERNAL_NET any

# Jalur direktori rules
var RULE_PATH /etc/snort/rules

# Menghubungkan ke file local.rules
include $RULE_PATH/local.rules
```

tambahan Edit file konfigurasi:

`sudo nano /etc/snort/rules/local.rules`

```
GNU nano 4.8 /etc/snort/rules/local.rules
alert icmp any any -> $HOME_NET any (msg:"Terdeteksi SERANGAN PING ICMP!"; sid:1000001; rev:1;)
alert tcp any any -> $HOME_NET any (msg:"Terdeteksi Potensi SYN FLOOD FLOODING!"; flags:S; sid:1000002; rev:1;)
```

isi ini :

`alert icmp any any -> $HOME_NET any (msg:"TERDETEKSI SERANGAN PING ICMP!"; sid:1000001; rev:1;)`

`alert tcp any any -> $HOME_NET any (msg:"TERDETEKSI POTENSI SYN FLOOD FLOODING!"; flags:S; sid:1000002; rev:1;)`

3. Menjalankan Snort

`sudo snort -A console -q -c /etc/snort/snort.conf -i eth0`

```
root@kevin:/home/kevin# sudo snort -A console -q -c /etc/snort/snort.conf -i ens33
-
```

Penjelasan:

- `-A console` → tampilkan alert di terminal
- `-q` → mode quiet
- `-c` → file konfigurasi
- `-i eth0` → interface jaringan

4. Cek IP Target

Cek IP target:

ip a

```
root@kevin:/home/kevin# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:2f:19:ae brd ff:ff:ff:ff:ff:ff
    inet 192.168.1.36/24 metric 100 brd 192.168.1.255 scope global dynamic ens33
        valid_lft 171460sec preferred_lft 171460sec
    inet6 fe80::20c:29ff:fe2f:19ae/64 scope link
        valid_lft forever preferred_lft forever
3: ens34: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:2f:19:b8 brd ff:ff:ff:ff:ff:ff
    inet 192.168.237.128/24 metric 100 brd 192.168.237.255 scope global dynamic ens34
        valid_lft 1360sec preferred_lft 1360sec
    inet6 fe80::20c:29ff:fe2f:19b8/64 scope link
        valid_lft forever preferred_lft forever
root@kevin:/home/kevin#
```

5. Simulasi Serangan

Terminal Attacker VM

Jalankan:

ping -f IP_TARGET

```
root@kevin:/home/kevin# ping -f 192.168.4.224
PING 192.168.4.224 (192.168.4.224) 56(84) bytes of data.
_
```

atau

ping -i 0.2 IP_TARGET (192.168.56.x)

```
root@kevin:/home/kevin# ping -i 0.2 192.168.4.224
PING 192.168.4.224 (192.168.4.224) 56(84) bytes of data.
_
```

- kirim tiap 0.2 detik (200ms)
- tidak terlalu agresif

6. Observasi Hasil

Amati:

- Alert yang muncul di terminal

05/19-12:48:00.720517 [**] [1:1000001:1] Terdeteksi SERANGAN
PING ICMP! [**] [Priority: 0] {ICMP} 192.168.4.240 -> 192.168.4.224

- Jenis serangan yang terdeteksi

serangan yang terdeteksi adalah ICMP Flooding (Ping Flood) atau
Potential Denial of Service (DoS) menggunakan protokol ICMP.

- Waktu dan log aktivitas

05/19, 12:48:00 hingga 12:48:01, Protokol: {ICMP}

Terjadi pertukaran paket dari IP Attacker (192.168.4.224) menuju IP
Target (192.168.4.240) dan sebaliknya.

Aktivitas pemantauan Snort dihentikan secara manual oleh pengguna
menggunakan perintah interupsi sistem ^C*** Caught Int-Signal

```
root@kevin:/home/kevin# sudo snort -A console -q -c /etc/snort/snort.conf -i ens33
05/19-12:48:00.720517 [**] [1:1000001:1] Terdeteksi SERANGAN PING ICMP! [**] [Priority: 0] {ICMP} 1
92.168.4.240 -> 192.168.4.224
05/19-12:48:00.720541 [**] [1:1000001:1] Terdeteksi SERANGAN PING ICMP! [**] [Priority: 0] {ICMP} 1
92.168.4.224 -> 192.168.4.240
05/19-12:48:00.921615 [**] [1:1000001:1] Terdeteksi SERANGAN PING ICMP! [**] [Priority: 0] {ICMP} 1
92.168.4.240 -> 192.168.4.224
05/19-12:48:00.921672 [**] [1:1000001:1] Terdeteksi SERANGAN PING ICMP! [**] [Priority: 0] {ICMP} 1
92.168.4.224 -> 192.168.4.240
05/19-12:48:01.123272 [**] [1:1000001:1] Terdeteksi SERANGAN PING ICMP! [**] [Priority: 0] {ICMP} 1
92.168.4.240 -> 192.168.4.224
05/19-12:48:01.123322 [**] [1:1000001:1] Terdeteksi SERANGAN PING ICMP! [**] [Priority: 0] {ICMP} 1
92.168.4.224 -> 192.168.4.240
05/19-12:48:01.326758 [**] [1:1000001:1] Terdeteksi SERANGAN PING ICMP! [**] [Priority: 0] {ICMP} 1
92.168.4.240 -> 192.168.4.224
05/19-12:48:01.326829 [**] [1:1000001:1] Terdeteksi SERANGAN PING ICMP! [**] [Priority: 0] {ICMP} 1
92.168.4.224 -> 192.168.4.240
05/19-12:48:01.527914 [**] [1:1000001:1] Terdeteksi SERANGAN PING ICMP! [**] [Priority: 0] {ICMP} 1
92.168.4.240 -> 192.168.4.224
05/19-12:48:01.527965 [**] [1:1000001:1] Terdeteksi SERANGAN PING ICMP! [**] [Priority: 0] {ICMP} 1
92.168.4.224 -> 192.168.4.240
^C*** Caught Int-Signal
root@kevin:/home/kevin#
```

7. OPSIONAL SYN FLOOD (vm attacker)

Install:

```
sudo apt install hping3 -y
```

```

root@kevin:/home/kevin# sudo apt install hping3
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following packages were automatically installed and are no longer required:
  libfwupdplugin1 libxmlb1
Use 'sudo apt autoremove' to remove them.
The following additional packages will be installed:
  libtc18.6
Suggested packages:
  tc18.6
The following NEW packages will be installed:
  hping3 libtc18.6
0 upgraded, 2 newly installed, 0 to remove and 0 not upgraded.
Need to get 1,009 kB of archives.
After this operation, 4,392 kB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://archive.ubuntu.com/ubuntu focal/main amd64 libtc18.6 amd64 8.6.10+dfsg-1 [902 kB]
Get:2 http://archive.ubuntu.com/ubuntu focal/universe amd64 hping3 amd64 3.a2.ds2-9 [107 kB]
Fetched 1,009 kB in 8s (133 kB/s)
Selecting previously unselected package libtc18.6:amd64.
(Reading database ... 122945 files and directories currently installed.)
Preparing to unpack .../libtc18.6_8.6.10+dfsg-1_amd64.deb ...
Unpacking libtc18.6:amd64 (8.6.10+dfsg-1) ...
Selecting previously unselected package hping3.
Preparing to unpack .../hping3_3.a2.ds2-9_amd64.deb ...
Unpacking hping3 (3.a2.ds2-9) ...
Setting up libtc18.6:amd64 (8.6.10+dfsg-1) ...
Setting up hping3 (3.a2.ds2-9) ...
Processing triggers for man-db (2.9.1-1) ...
Processing triggers for libc-bin (2.31-0ubuntu9.18) ...
root@kevin:/home/kevin#

```

Jalankan:

`sudo hping3 -S --flood -p 80 IP_TARGET`

sisi attacker

```

root@kevin:/home/kevin# sudo hping3 -S --flood -p 80 192.168.4.224
HPING 192.168.4.224 (ens33 192.168.4.224): S set, 40 headers + 0 data bytes
hping in flood mode, no replies will be shown

```

sisi target

```

05/19-13:08:22.885848  [**] [1:1000002:1] Terdeteksi Potensi SYN FLOOD FLOODING! [**] [Priority: 0]
{TCP} 192.168.4.240:42165 -> 192.168.4.224:80
05/19-13:08:22.885848  [**] [1:1000002:1] Terdeteksi Potensi SYN FLOOD FLOODING! [**] [Priority: 0]
{TCP} 192.168.4.240:42166 -> 192.168.4.224:80
05/19-13:08:22.885848  [**] [1:1000002:1] Terdeteksi Potensi SYN FLOOD FLOODING! [**] [Priority: 0]
{TCP} 192.168.4.240:42167 -> 192.168.4.224:80
05/19-13:08:22.885848  [**] [1:1000002:1] Terdeteksi Potensi SYN FLOOD FLOODING! [**] [Priority: 0]
{TCP} 192.168.4.240:42168 -> 192.168.4.224:80
05/19-13:08:22.885848  [**] [1:1000002:1] Terdeteksi Potensi SYN FLOOD FLOODING! [**] [Priority: 0]
{TCP} 192.168.4.240:42169 -> 192.168.4.224:80
05/19-13:08:22.886395  [**] [1:1000002:1] Terdeteksi Potensi SYN FLOOD FLOODING! [**] [Priority: 0]
{TCP} 192.168.4.240:42170 -> 192.168.4.224:80
05/19-13:08:22.886395  [**] [1:1000002:1] Terdeteksi Potensi SYN FLOOD FLOODING! [**] [Priority: 0]
{TCP} 192.168.4.240:42171 -> 192.168.4.224:80
05/19-13:08:22.886395  [**] [1:1000002:1] Terdeteksi Potensi SYN FLOOD FLOODING! [**] [Priority: 0]
{TCP} 192.168.4.240:42172 -> 192.168.4.224:80
05/19-13:08:22.886395  [**] [1:1000002:1] Terdeteksi Potensi SYN FLOOD FLOODING! [**] [Priority: 0]
{TCP} 192.168.4.240:42173 -> 192.168.4.224:80
05/19-13:08:22.886395  [**] [1:1000002:1] Terdeteksi Potensi SYN FLOOD FLOODING! [**] [Priority: 0]
{TCP} 192.168.4.240:42174 -> 192.168.4.224:80
05/19-13:08:22.886395  [**] [1:1000002:1] Terdeteksi Potensi SYN FLOOD FLOODING! [**] [Priority: 0]
{TCP} 192.168.4.240:42175 -> 192.168.4.224:80
05/19-13:08:22.886395  [**] [1:1000002:1] Terdeteksi Potensi SYN FLOOD FLOODING! [**] [Priority: 0]
{TCP} 192.168.4.240:42176 -> 192.168.4.224:80
05/19-13:08:22.886395  [**] [1:1000002:1] Terdeteksi Potensi SYN FLOOD FLOODING! [**] [Priority: 0]
{TCP} 192.168.4.240:42177 -> 192.168.4.224:80
^C*** Caught Int-Signal
root@kevin:/home/kevin# _

```


- Catatan:
Ini adalah simulasi yang lebih realistis menggunakan SYN flood, yang sering digunakan dalam serangan DDoS nyata.

V. HASIL PRAKTIKUM

Tuliskan hasil:

- IDS berhasil berjalan / tidak
- Alert yang muncul
- Jenis aktivitas yang terdeteksi
- Screenshot (lampirkan)

VI. ANALISIS

Jawab pertanyaan berikut:

1. Apa fungsi utama IDS dalam sistem keamanan?

jawab : Fungsi utama IDS adalah sebagai sistem pengawasan (monitoring dan auditing) yang bertugas mendeteksi aktivitas mencurigakan atau serangan yang sedang terjadi pada jaringan. IDS juga berfungsi memberikan peringatan (*alert*) kepada administrator jaringan agar tindakan mitigasi bisa segera diambil sebelum kerusakan meluas.

2. Bagaimana cara kerja signature-based IDS?

jawab : Signature-based IDS bekerja mirip seperti perangkat lunak antivirus. Sistem ini mencocokkan setiap paket data atau lalu lintas jaringan yang lewat dengan pola serangan yang sudah diketahui sebelumnya. Jika ada paket data yang memiliki pola atau karakteristik yang persis sama dengan salah satu serangan (misalnya aturan deteksi ICMP atau string teks serangan eksploit tertentu), Snort akan langsung memicu alert.

3. Mengapa anomaly-based IDS bisa menghasilkan false positive?

jawab : Anomaly-based IDS bekerja dengan cara membuat profil perilaku jaringan yang dianggap normal terlebih dahulu. Sistem akan menganggap

aktivitas apa pun yang keluar dari batas tersebut sebagai serangan. Hal ini sering memicu false positive (aktivitas normal dianggap serangan) karena perilaku pengguna asli atau aplikasi di jaringan nyata sangat dinamis. Misalnya, ketika ada pembaruan perangkat lunak otomatis atau transfer file berukuran besar yang sah, IDS mendeteksinya sebagai lonjakan trafik padahal itu bukan serangan.

4. Apa perbedaan IDS dan IPS dalam praktik nyata?

jawab : IDS hanya memberikan log/alert di konsol dan membiarkan paket berbahaya tersebut tetap lewat ke target. Sementara IPS akan langsung memblokir paket tersebut atau langsung memberikan tindakan (misalnya menjatuhkan koneksi TCP atau memerintahkan firewall menutup IP penyerang secara otomatis).

5. Apakah IDS cukup untuk melindungi sistem? Jelaskan!

jawab : Tidak cukup. IDS hanyalah salah satu komponen dalam strategi keamanan berlapis. Karena IDS bersifat pasif. Jika admin tidak siaga melihat alert, serangan akan tetap berhasil menembus target karena IDS tidak melakukan pencegahan secara instan. kemudian Signature-based IDS tidak berdaya melawan serangan baru yang polanya belum terdaftar.

Oleh karena itu, IDS harus dikombinasikan dengan perangkat keamanan lain seperti Firewall, IPS, Antivirus, enkripsi data, dan pembaruan patch keamanan sistem secara berkala.